

José Carter Arriagada

Software Engineer | Production Technology, Pipeline Tools & Simulation

Santiago, Chile | Chile: +56 9 8655 2408 | U.S.: +1 276 221 3064

jose.carterx@gmail.com | josecarter.dev/visual

linkedin.com/in/jose-carter-arriagada | github.com/Cartterr

Profile

Software engineer building production platforms, workflow automation, data pipelines, and simulation tools with Python and TypeScript. ACM SIGGRAPH 2026 Student Volunteer focused on production technology, pipeline tooling, and artist-facing animation/VFX systems.

Technical Skills

Languages: Python, TypeScript/JavaScript, C++, Java, SQL **Production:** Node.js, Angular/Nx, PostgreSQL, AWS, Docker, Linux, Git, CI/CD

Visual & Compute: Three.js, WebGL, CUDA, Houdini, Cinema 4D, parametric geometry, scene graphs, scientific visualization, simulation, model export

Specialties: Production tooling, workflow automation, distributed/event-driven systems, data processing, performance optimization, AI/LLM integrations **English:** Advanced (TOEIC 905)

Selected Experience

Software Engineer

Sep 2025 – Present

Dily - Automotive Fintech Platform
Santiago, Chile

- Build end-to-end lending and automotive-finance workflows across **TypeScript/Node.js** services, **Angular/Nx** micro-frontends, PostgreSQL/MySQL, and AWS integrations.
- Apply domain-driven and hexagonal architecture with automated backend and frontend testing to keep production systems modular, reliable, and maintainable.

Data Science Researcher - Platform Engineering

Jul 2023 – Jul 2025

Pontificia Universidad Católica de Chile - Politiktok (Fondecyt)
Santiago, Chile

- Lead engineering of a **Python/PostgreSQL** research platform processing **100,000+ records**; achieved approximately **10x faster processing** through parallelization and query optimization.
- Built data-ingestion, NLP, API, and visualization workflows supporting multidisciplinary research and a peer-reviewed Springer publication.

Software Engineer (Full Stack)

Dec 2024 – Aug 2025

Flair - Energy Optimization Platform
Santiago, Chile

- Developed Python microservices and Vue/React applications deployed on AWS with Docker; integrated InfluxDB and DynamoDB for high-frequency energy-data workflows.

Software Engineer - Drone Response

Jan 2024 – Mar 2024

University of Notre Dame - International Research Internship
Notre Dame, IN, USA

- Developed a smart mission planner for multi-drone coordination using Python route optimization, MQTT event-driven communication, OpenAI API integration, and Java Spring services.

Teaching & Technical Mentoring

Mar 2023 – Jul 2026

Escuela Militar (Jul 2024-Jul 2026) | PUC DCC (Mar 2023-Jul 2025)
Santiago, Chile

- Taught Python, authored assessments, and advised cadets; completed **14+ assistantships** across Operating Systems, Networks, HPC, Extended Reality, and Testing; received 2024/2026 teaching recognition and was named **Senior Teaching Assistant** in 2025.

Selected Visual & Production Tools

Parametric 3D Configurator

2026

Independent product-visualization tool - TypeScript, Three.js, Vite, Playwright

- Built interactive plan and 3D editing views that convert room and shelving measurements into parametric scene geometry, physical-constraint feedback, and export-oriented workflows.

Simulation Engineer

Aug 2022 – Dec 2024

Pontificia Universidad Católica de Chile - Geoscience and 3D visualization
Santiago, Chile

- Built tectonic-fault simulation and spatial-visualization tooling using Python, CUDA, Tectosaur, Mapbox, and Houdini/Cinema 4D workflows, paired with up to a **15x simulation speedup**.

Research, Community & Recognition

ACM SIGGRAPH 2026 Student Volunteer — Supported conference operations in Los Angeles and attended sessions on production pipelines, rendering, real-time systems, XR, AI, and studio technology.

Publications — Co-author: *Political Subjectification on TikTok: A Postdigital Mode of Inquiry for Tracing Affects Among Chilean Youth* (Springer, 2026); *Interactions Between Megathrust and Adjacent Crustal Faults* (SSA, 2023).

Professional Development — SGAC 10th South American Space Generation Workshop Delegate (2026); KHIPU 2025 - Latin American Meeting in Artificial Intelligence (Credential 572/29865).

Education

Pontificia Universidad Católica de Chile

Mar 2020 – Jul 2025

B.S. in Computer Science Engineering; Major in Software Engineering; Minor in Data Science

Santiago, Chile

Graduated with Distinction